numpy

25th

In [1]:
```
import numpy as np
```

In [2]:
```
np.__version__
```

Out[2]:
```
'2.3.5'
```

In [3]:
```
my_list = [0,1,2,3,4,5]
```

In [4]:
```
my_list
```

Out[4]:
```
[0, 1, 2, 3, 4, 5]
```

In [5]:
```
type(my_list)
```

Out[5]:
```
list
```

In [6]:
```
arr = np.array(my_list)
arr
```

Out[6]:
```
array([0, 1, 2, 3, 4, 5])
```

In [7]:
```
type(arr)
```

Out[7]:
```
numpy.ndarray
```

In [8]:
```
arr
```

Out[8]:
```
array([0, 1, 2, 3, 4, 5])
```

In [9]:
```
np.arange(10)
```

Out[9]:
```
array([0, 1, 2, 3, 4, 5, 6, 7, 8, 9])
```

In [10]:
```
np.arange(20)
```

Out[10]:
```
array([ 0,  1,  2,  3,  4,  5,  6,  7,  8,  9, 10, 11, 12, 13, 14, 15, 16,
       17, 18, 19])
```

In [11]:
```
np.arange(10,20,3)
```

Out[11]:
```
array([10, 13, 16, 19])
```

In [12]:
```
np.zeros(5)
```

Out[12]:
```
array([0., 0., 0., 0., 0.])
```

In [13]:
```
np.zeros(5,dtype=int)
```

Out[13]:
```
array([0, 0, 0, 0, 0])
```

In [14]:
```
np.zeros((2,2))   # 2nd array
```

Out[14]:
```
array([[0., 0.],
       [0., 0.]])
```

In [15]:
```
np.zeros((2,2),dtype=int)
```

Out[15]:
```
array([[0, 0],
       [0, 0]])
```

In [16]:
```
np.zeros((5,10),dtype=int)
```

Out[16]:
```
array([[0, 0, 0, 0, 0, 0, 0, 0, 0, 0],
       [0, 0, 0, 0, 0, 0, 0, 0, 0, 0],
       [0, 0, 0, 0, 0, 0, 0, 0, 0, 0],
       [0, 0, 0, 0, 0, 0, 0, 0, 0, 0],
       [0, 0, 0, 0, 0, 0, 0, 0, 0, 0]])
```

In [17]:
```
np.ones(10)
```

Out[17]:
```
array([1., 1., 1., 1., 1., 1., 1., 1., 1., 1.])
```

In [18]:
```
np.ones((3,3),dtype=int)
```

Out[18]:
```
array([[1, 1, 1],
       [1, 1, 1],
       [1, 1, 1]])
```

In [19]:
```
np.twos(3)
```

```
---------------------------------------------------------------------------
AttributeError                            Traceback (most recent call last)
Cell In[19], line 1
----> 1 np.twos(3)

File ~\anaconda3\Lib\site-packages\numpy\__init__.py:795, in __getattr__(attr)
    792     import numpy.char as char
    793     return char.chararray
--> 795 raise AttributeError(f"module {__name__!r} has no attribute {attr!r}")

AttributeError: module 'numpy' has no attribute 'twos'
```

In [20]:
```
np.fours(3)
```

```
---------------------------------------------------------------------------
AttributeError                            Traceback (most recent call last)
Cell In[20], line 1
----> 1 np.fours(3)

File ~\anaconda3\Lib\site-packages\numpy\__init__.py:795, in __getattr__(attr)
    792     import numpy.char as char
    793     return char.chararray
--> 795 raise AttributeError(f"module {__name__!r} has no attribute {attr!r}")

AttributeError: module 'numpy' has no attribute 'fours'
```

In [21]:
```
rand(4)
```

```
---------------------------------------------------------------------------
NameError                                 Traceback (most recent call last)
Cell In[21], line 1
----> 1 rand(4)

NameError: name 'rand' is not defined
```

In [22]:
```
random.rand(4)
```

```
---------------------------------------------------------------------------
NameError                                 Traceback (most recent call last)
Cell In[22], line 1
----> 1 random.rand(4)

NameError: name 'random' is not defined
```

In [23]:
```
np.random.rand(4)
```

Out[23]:
```
array([0.15101357, 0.27134375, 0.12010334, 0.51196369])
```

In [24]:
```
np.random.rand(4)
```

Out[24]:
```
array([0.5217521 , 0.09635599, 0.36603116, 0.65638353])
```

In [25]:
```
np.random.rand(2,2)
```

Out[25]:
```
array([[0.05354703, 0.57790939],
       [0.16512957, 0.28055244]])
```

In [26]:
```
np.random.randint(3)
```

Out[26]:
```
2
```

In [27]:
```
np.random.randint(3)
```

Out[27]:
```
2
```

In [28]:
```
np.random.randint(0,3)
```

Out[28]:
```
2
```

In [29]:
```
np.random.randint(0,3)
```

Out[29]:
```
1
```

In [30]:
```
np.random.randint(10,15)
```

Out[30]:
```
13
```

In [31]:
```
np.random.randint(10,15)
```

Out[31]:
```
14
```

In [32]:
```
np.random.randint(10,21,3)
```

Out[32]:
```
array([19, 15, 10], dtype=int32)
```

In [33]:
```
np.random.randint(1,12,10)
```

Out[33]:
```
array([ 5,  2,  4,  3, 10,  9, 11,  9,  4,  8], dtype=int32)
```

In [34]:
```
np.random.randint(10,40,(10,10))
```

Out[34]:
```
array([[33, 19, 12, 32, 35, 34, 10, 19, 35, 29],
       [38, 35, 18, 24, 30, 34, 28, 32, 27, 21],
       [39, 16, 21, 39, 10, 17, 33, 19, 14, 33],
       [22, 31, 36, 37, 17, 31, 37, 24, 23, 33],
       [28, 17, 34, 36, 26, 36, 20, 32, 12, 10],
       [11, 19, 20, 25, 23, 15, 16, 31, 22, 14],
       [34, 29, 12, 33, 18, 33, 18, 37, 25, 35],
       [22, 33, 30, 20, 36, 30, 38, 11, 20, 28],
       [36, 21, 27, 18, 11, 12, 31, 37, 30, 34],
       [37, 18, 14, 11, 27, 34, 24, 11, 26, 38]], dtype=int32)
```

In [35]:
```
np.random.randint(10,40,(4,5))
```

Out[35]:
```
array([[24, 11, 34, 35, 15],
       [28, 38, 15, 17, 23],
       [23, 33, 24, 25, 25],
       [24, 25, 13, 22, 25]], dtype=int32)
```

In [36]:
```
np.random.randint(0,100,(12,12))
```

Out[36]:
```
array([[27, 56, 21, 56, 17, 83, 50, 56, 37, 25, 58, 38],
       [46, 46, 41, 80,  6, 81, 57, 49, 21, 19, 65,  1],
       [95,  5, 64, 97, 99, 82, 50, 71, 15, 30, 20, 51],
       [41, 49, 19, 60, 73, 39,  3, 47, 59, 23,  4, 55],
       [88,  5, 61, 10, 94, 83, 64, 92, 57, 40, 34, 75],
       [84, 69, 37, 12, 65, 88, 38, 73, 94, 55,  6, 30],
       [28, 45, 63, 91, 79, 68, 14, 30, 88, 83, 97, 83],
       [74,  8, 37, 13, 95, 36,  3, 69, 77, 64, 18, 89],
       [41, 38, 72, 17, 81, 83, 49, 17, 35, 97, 38, 54],
       [79, 94, 52, 46, 15, 70, 60, 48, 92, 62, 39, 63],
       [70, 39,  6,  3, 14, 45, 87, 97, 83,  4,  2, 57],
       [98, 92, 41, 67, 61,  1, 48, 25, 88,  1, 52, 16]], dtype=int32)
```

In [37]:
```
arr = np.random.randint(10,40,(10,10))
```

In [38]:
```
arr
```

Out[38]:
```
array([[32, 34, 27, 35, 19, 18, 23, 23, 31, 38],
       [38, 34, 27, 33, 26, 39, 22, 38, 21, 29],
       [17, 38, 37, 19, 17, 16, 13, 38, 17, 38],
       [23, 33, 16, 21, 34, 14, 32, 10, 34, 38],
       [13, 32, 25, 12, 38, 16, 23, 38, 10, 27],
       [11, 28, 31, 14, 22, 33, 39, 28, 23, 37],
       [29, 34, 39, 13, 20, 36, 11, 32, 10, 35],
       [28, 28, 10, 19, 19, 32, 23, 25, 28, 23],
       [17, 30, 36, 18, 14, 31, 12, 25, 35, 12],
       [33, 33, 30, 20, 20, 25, 30, 29, 38, 25]], dtype=int32)
```

In [39]:
```
id(arr)
```

Out[39]:
```
2205144439024
```

In [40]:
```
type(arr)
```

Out[40]:
```
numpy.ndarray
```

In [41]:
```
arr[:]
```

Out[41]:
```
array([[32, 34, 27, 35, 19, 18, 23, 23, 31, 38],
       [38, 34, 27, 33, 26, 39, 22, 38, 21, 29],
       [17, 38, 37, 19, 17, 16, 13, 38, 17, 38],
       [23, 33, 16, 21, 34, 14, 32, 10, 34, 38],
       [13, 32, 25, 12, 38, 16, 23, 38, 10, 27],
       [11, 28, 31, 14, 22, 33, 39, 28, 23, 37],
       [29, 34, 39, 13, 20, 36, 11, 32, 10, 35],
       [28, 28, 10, 19, 19, 32, 23, 25, 28, 23],
       [17, 30, 36, 18, 14, 31, 12, 25, 35, 12],
       [33, 33, 30, 20, 20, 25, 30, 29, 38, 25]], dtype=int32)
```

In [42]:
```
arr[2:5]
```

Out[42]:
```
array([[17, 38, 37, 19, 17, 16, 13, 38, 17, 38],
       [23, 33, 16, 21, 34, 14, 32, 10, 34, 38],
       [13, 32, 25, 12, 38, 16, 23, 38, 10, 27]], dtype=int32)
```

In [43]:
```
arr[6:9]
```

Out[43]:
```
array([[29, 34, 39, 13, 20, 36, 11, 32, 10, 35],
       [28, 28, 10, 19, 19, 32, 23, 25, 28, 23],
       [17, 30, 36, 18, 14, 31, 12, 25, 35, 12]], dtype=int32)
```

In [48]:
```
arr
```

Out[48]:
```
array([[32, 34, 27, 35, 19, 18, 23, 23, 31, 38],
       [38, 34, 27, 33, 26, 39, 22, 38, 21, 29],
       [17, 38, 37, 19, 17, 16, 13, 38, 17, 38],
       [23, 33, 16, 21, 34, 14, 32, 10, 34, 38],
       [13, 32, 25, 12, 38, 16, 23, 38, 10, 27],
       [11, 28, 31, 14, 22, 33, 39, 28, 23, 37],
       [29, 34, 39, 13, 20, 36, 11, 32, 10, 35],
       [28, 28, 10, 19, 19, 32, 23, 25, 28, 23],
       [17, 30, 36, 18, 14, 31, 12, 25, 35, 12],
       [33, 33, 30, 20, 20, 25, 30, 29, 38, 25]], dtype=int32)
```

In [58]:
```
arr[5,7]
```

Out[58]:
```
np.int32(28)
```

In [59]:
```
arr[3,6]
```

Out[59]:
```
np.int32(32)
```

In [62]:
```
arr
```

Out[62]:
```
array([[32, 34, 27, 35, 19, 18, 23, 23, 31, 38],
       [38, 34, 27, 33, 26, 39, 22, 38, 21, 29],
       [17, 38, 37, 19, 17, 16, 13, 38, 17, 38],
       [23, 33, 16, 21, 34, 14, 32, 10, 34, 38],
       [13, 32, 25, 12, 38, 16, 23, 38, 10, 27],
       [11, 28, 31, 14, 22, 33, 39, 28, 23, 37],
       [29, 34, 39, 13, 20, 36, 11, 32, 10, 35],
       [28, 28, 10, 19, 19, 32, 23, 25, 28, 23],
       [17, 30, 36, 18, 14, 31, 12, 25, 35, 12],
       [33, 33, 30, 20, 20, 25, 30, 29, 38, 25]], dtype=int32)
```

In [63]:
```
arr<30
```

Out[63]:
```
array([[False, False,  True, False,  True,  True,  True,  True, False,
        False],
       [False, False,  True, False,  True, False,  True, False,  True,
         True],
       [ True, False, False,  True,  True,  True,  True, False,  True,
        False],
       [ True, False,  True,  True, False,  True, False,  True, False,
        False],
       [ True, False,  True,  True, False,  True,  True, False,  True,
         True],
       [ True,  True, False,  True,  True, False, False,  True,  True,
        False],
       [ True, False, False,  True,  True, False,  True, False,  True,
        False],
       [ True,  True,  True,  True,  True, False,  True,  True,  True,
         True],
       [ True, False, False,  True,  True, False,  True,  True, False,
         True],
       [False, False, False,  True,  True,  True, False,  True, False,
         True]])
```

In [64]:
```
arr[arr<30]
```

Out[64]:
```
array([27, 19, 18, 23, 23, 27, 26, 22, 21, 29, 17, 19, 17, 16, 13, 17, 23,
       16, 21, 14, 10, 13, 25, 12, 16, 23, 10, 27, 11, 28, 14, 22, 28, 23,
       29, 13, 20, 11, 10, 28, 28, 10, 19, 19, 23, 25, 28, 23, 17, 18, 14,
       12, 25, 12, 20, 20, 25, 29, 25], dtype=int32)
```

In [65]:
```
arr[arr<=30]
```

Out[65]:
```
array([27, 19, 18, 23, 23, 27, 26, 22, 21, 29, 17, 19, 17, 16, 13, 17, 23,
       16, 21, 14, 10, 13, 25, 12, 16, 23, 10, 27, 11, 28, 14, 22, 28, 23,
       29, 13, 20, 11, 10, 28, 28, 10, 19, 19, 23, 25, 28, 23, 17, 30, 18,
       14, 12, 25, 12, 30, 20, 20, 25, 30, 29, 25], dtype=int32)
```

In [66]:
```
arr[arr>30]
```

Out[66]:
```
array([32, 34, 35, 31, 38, 38, 34, 33, 39, 38, 38, 37, 38, 38, 33, 34, 32,
       34, 38, 32, 38, 38, 31, 33, 39, 37, 34, 39, 36, 32, 35, 32, 36, 31,
       35, 33, 33, 38], dtype=int32)
```

In [67]:
```
arr[arr>=32]
```

Out[67]:
```
array([32, 34, 35, 38, 38, 34, 33, 39, 38, 38, 37, 38, 38, 33, 34, 32, 34,
       38, 32, 38, 38, 33, 39, 37, 34, 39, 36, 32, 35, 32, 36, 35, 33, 33,
       38], dtype=int32)
```

In [68]:
```
arr[arr==30]
```

Out[68]:
```
array([30, 30, 30], dtype=int32)
```

In [69]:
```
arr[arr==22]
```

Out[69]:
```
array([22, 22], dtype=int32)
```

In [70]:
```
arr[arr==38]
```

Out[70]:
```
array([38, 38, 38, 38, 38, 38, 38, 38, 38, 38], dtype=int32)
```

In [71]:
```
arr
```

Out[71]:
```
array([[32, 34, 27, 35, 19, 18, 23, 23, 31, 38],
       [38, 34, 27, 33, 26, 39, 22, 38, 21, 29],
       [17, 38, 37, 19, 17, 16, 13, 38, 17, 38],
       [23, 33, 16, 21, 34, 14, 32, 10, 34, 38],
       [13, 32, 25, 12, 38, 16, 23, 38, 10, 27],
       [11, 28, 31, 14, 22, 33, 39, 28, 23, 37],
       [29, 34, 39, 13, 20, 36, 11, 32, 10, 35],
       [28, 28, 10, 19, 19, 32, 23, 25, 28, 23],
       [17, 30, 36, 18, 14, 31, 12, 25, 35, 12],
       [33, 33, 30, 20, 20, 25, 30, 29, 38, 25]], dtype=int32)
```

In [72]:
```
arr[3:5,4:6]
```

Out[72]:
```
array([[34, 14],
       [38, 16]], dtype=int32)
```

In [73]:
```
arr[6:8,4:6]
```

Out[73]:
```
array([[20, 36],
       [19, 32]], dtype=int32)
```

In [74]:
```
arr[4:5,4:5]
```

Out[74]:
```
array([[38]], dtype=int32)
```

In [75]:
```
arr[5:6,5:6]
```

Out[75]:
```
array([[33]], dtype=int32)
```

In [77]:
```
arr
```

Out[77]:
```
array([[32, 34, 27, 35, 19, 18, 23, 23, 31, 38],
       [38, 34, 27, 33, 26, 39, 22, 38, 21, 29],
       [17, 38, 37, 19, 17, 16, 13, 38, 17, 38],
       [23, 33, 16, 21, 34, 14, 32, 10, 34, 38],
       [13, 32, 25, 12, 38, 16, 23, 38, 10, 27],
       [11, 28, 31, 14, 22, 33, 39, 28, 23, 37],
       [29, 34, 39, 13, 20, 36, 11, 32, 10, 35],
       [28, 28, 10, 19, 19, 32, 23, 25, 28, 23],
       [17, 30, 36, 18, 14, 31, 12, 25, 35, 12],
       [33, 33, 30, 20, 20, 25, 30, 29, 38, 25]], dtype=int32)
```

In [78]:
```
arr[:-1]
```

Out[78]:
```
array([[32, 34, 27, 35, 19, 18, 23, 23, 31, 38],
       [38, 34, 27, 33, 26, 39, 22, 38, 21, 29],
       [17, 38, 37, 19, 17, 16, 13, 38, 17, 38],
       [23, 33, 16, 21, 34, 14, 32, 10, 34, 38],
       [13, 32, 25, 12, 38, 16, 23, 38, 10, 27],
       [11, 28, 31, 14, 22, 33, 39, 28, 23, 37],
       [29, 34, 39, 13, 20, 36, 11, 32, 10, 35],
       [28, 28, 10, 19, 19, 32, 23, 25, 28, 23],
       [17, 30, 36, 18, 14, 31, 12, 25, 35, 12]], dtype=int32)
```

In [79]:
```
arr
```

Out[79]:
```
array([[32, 34, 27, 35, 19, 18, 23, 23, 31, 38],
       [38, 34, 27, 33, 26, 39, 22, 38, 21, 29],
       [17, 38, 37, 19, 17, 16, 13, 38, 17, 38],
       [23, 33, 16, 21, 34, 14, 32, 10, 34, 38],
       [13, 32, 25, 12, 38, 16, 23, 38, 10, 27],
       [11, 28, 31, 14, 22, 33, 39, 28, 23, 37],
       [29, 34, 39, 13, 20, 36, 11, 32, 10, 35],
       [28, 28, 10, 19, 19, 32, 23, 25, 28, 23],
       [17, 30, 36, 18, 14, 31, 12, 25, 35, 12],
       [33, 33, 30, 20, 20, 25, 30, 29, 38, 25]], dtype=int32)
```

In [80]:
```
arr[3:-4]
```

Out[80]:
```
array([[23, 33, 16, 21, 34, 14, 32, 10, 34, 38],
       [13, 32, 25, 12, 38, 16, 23, 38, 10, 27],
       [11, 28, 31, 14, 22, 33, 39, 28, 23, 37]], dtype=int32)
```

In [82]:
```
arr
```

Out[82]:
```
array([[32, 34, 27, 35, 19, 18, 23, 23, 31, 38],
       [38, 34, 27, 33, 26, 39, 22, 38, 21, 29],
       [17, 38, 37, 19, 17, 16, 13, 38, 17, 38],
       [23, 33, 16, 21, 34, 14, 32, 10, 34, 38],
       [13, 32, 25, 12, 38, 16, 23, 38, 10, 27],
       [11, 28, 31, 14, 22, 33, 39, 28, 23, 37],
       [29, 34, 39, 13, 20, 36, 11, 32, 10, 35],
       [28, 28, 10, 19, 19, 32, 23, 25, 28, 23],
       [17, 30, 36, 18, 14, 31, 12, 25, 35, 12],
       [33, 33, 30, 20, 20, 25, 30, 29, 38, 25]], dtype=int32)
```

In [84]:
```
arr[5:-2]
```

Out[84]:
```
array([[11, 28, 31, 14, 22, 33, 39, 28, 23, 37],
       [29, 34, 39, 13, 20, 36, 11, 32, 10, 35],
       [28, 28, 10, 19, 19, 32, 23, 25, 28, 23]], dtype=int32)
```

In [85]:
```
arr[1:4]
```

Out[85]:
```
array([[38, 34, 27, 33, 26, 39, 22, 38, 21, 29],
       [17, 38, 37, 19, 17, 16, 13, 38, 17, 38],
       [23, 33, 16, 21, 34, 14, 32, 10, 34, 38]], dtype=int32)
```

In [86]:
```
arr[6]
```

Out[86]:
```
array([29, 34, 39, 13, 20, 36, 11, 32, 10, 35], dtype=int32)
```

In [87]:
```
arr[0]
```

Out[87]:
```
array([32, 34, 27, 35, 19, 18, 23, 23, 31, 38], dtype=int32)
```

In [88]:
```
arr[5:]
```

Out[88]:
```
array([[11, 28, 31, 14, 22, 33, 39, 28, 23, 37],
       [29, 34, 39, 13, 20, 36, 11, 32, 10, 35],
       [28, 28, 10, 19, 19, 32, 23, 25, 28, 23],
       [17, 30, 36, 18, 14, 31, 12, 25, 35, 12],
       [33, 33, 30, 20, 20, 25, 30, 29, 38, 25]], dtype=int32)
```

In [89]:
```
arr[:5]
```

Out[89]:
```
array([[32, 34, 27, 35, 19, 18, 23, 23, 31, 38],
       [38, 34, 27, 33, 26, 39, 22, 38, 21, 29],
       [17, 38, 37, 19, 17, 16, 13, 38, 17, 38],
       [23, 33, 16, 21, 34, 14, 32, 10, 34, 38],
       [13, 32, 25, 12, 38, 16, 23, 38, 10, 27]], dtype=int32)
```

In [91]:
```
arr[4:6]
```

Out[91]:
```
array([[13, 32, 25, 12, 38, 16, 23, 38, 10, 27],
       [11, 28, 31, 14, 22, 33, 39, 28, 23, 37]], dtype=int32)
```

In [92]:
```
arr
```

Out[92]:
```
array([[32, 34, 27, 35, 19, 18, 23, 23, 31, 38],
       [38, 34, 27, 33, 26, 39, 22, 38, 21, 29],
       [17, 38, 37, 19, 17, 16, 13, 38, 17, 38],
       [23, 33, 16, 21, 34, 14, 32, 10, 34, 38],
       [13, 32, 25, 12, 38, 16, 23, 38, 10, 27],
       [11, 28, 31, 14, 22, 33, 39, 28, 23, 37],
       [29, 34, 39, 13, 20, 36, 11, 32, 10, 35],
       [28, 28, 10, 19, 19, 32, 23, 25, 28, 23],
       [17, 30, 36, 18, 14, 31, 12, 25, 35, 12],
       [33, 33, 30, 20, 20, 25, 30, 29, 38, 25]], dtype=int32)
```

In [93]:
```
arr[2:6:3]
```

Out[93]:
```
array([[17, 38, 37, 19, 17, 16, 13, 38, 17, 38],
       [11, 28, 31, 14, 22, 33, 39, 28, 23, 37]], dtype=int32)
```

In [94]:
```
arr[2:8:2]
```

Out[94]:
```
array([[17, 38, 37, 19, 17, 16, 13, 38, 17, 38],
       [13, 32, 25, 12, 38, 16, 23, 38, 10, 27],
       [29, 34, 39, 13, 20, 36, 11, 32, 10, 35]], dtype=int32)
```

In [95]:
```
arr
```

Out[95]:
```
array([[32, 34, 27, 35, 19, 18, 23, 23, 31, 38],
       [38, 34, 27, 33, 26, 39, 22, 38, 21, 29],
       [17, 38, 37, 19, 17, 16, 13, 38, 17, 38],
       [23, 33, 16, 21, 34, 14, 32, 10, 34, 38],
       [13, 32, 25, 12, 38, 16, 23, 38, 10, 27],
       [11, 28, 31, 14, 22, 33, 39, 28, 23, 37],
       [29, 34, 39, 13, 20, 36, 11, 32, 10, 35],
       [28, 28, 10, 19, 19, 32, 23, 25, 28, 23],
       [17, 30, 36, 18, 14, 31, 12, 25, 35, 12],
       [33, 33, 30, 20, 20, 25, 30, 29, 38, 25]], dtype=int32)
```

In [96]:
```
arr[::-3]
```

Out[96]:
```
array([[33, 33, 30, 20, 20, 25, 30, 29, 38, 25],
       [29, 34, 39, 13, 20, 36, 11, 32, 10, 35],
       [23, 33, 16, 21, 34, 14, 32, 10, 34, 38],
       [32, 34, 27, 35, 19, 18, 23, 23, 31, 38]], dtype=int32)
```

In [97]:
```
arr[::3]
```

Out[97]:
```
array([[32, 34, 27, 35, 19, 18, 23, 23, 31, 38],
       [23, 33, 16, 21, 34, 14, 32, 10, 34, 38],
       [29, 34, 39, 13, 20, 36, 11, 32, 10, 35],
       [33, 33, 30, 20, 20, 25, 30, 29, 38, 25]], dtype=int32)
```

In [98]:
```
arr[::-5]
```

Out[98]:
```
array([[33, 33, 30, 20, 20, 25, 30, 29, 38, 25],
       [13, 32, 25, 12, 38, 16, 23, 38, 10, 27]], dtype=int32)
```

In [99]:
```
arr
```

Out[99]:
```
array([[32, 34, 27, 35, 19, 18, 23, 23, 31, 38],
       [38, 34, 27, 33, 26, 39, 22, 38, 21, 29],
       [17, 38, 37, 19, 17, 16, 13, 38, 17, 38],
       [23, 33, 16, 21, 34, 14, 32, 10, 34, 38],
       [13, 32, 25, 12, 38, 16, 23, 38, 10, 27],
       [11, 28, 31, 14, 22, 33, 39, 28, 23, 37],
       [29, 34, 39, 13, 20, 36, 11, 32, 10, 35],
       [28, 28, 10, 19, 19, 32, 23, 25, 28, 23],
       [17, 30, 36, 18, 14, 31, 12, 25, 35, 12],
       [33, 33, 30, 20, 20, 25, 30, 29, 38, 25]], dtype=int32)
```

In [100…
```
arr[4,7]
```

Out[100…
```
np.int32(38)
```

In [ ]:
```

```